

1 Benchmark Scope

Physical Domains

Translational Motion & Collisions



Rolling, Friction & Rigid-Body Statics



Pendulum Motion & Oscillations



Optics & Projective Geometry



Hydrostatics & Buoyancy



Phase Transitions & Melting



Electrostatics & Electromagnetism



Granular Media & Discharge Flow



Surface Tension & Viscous Flow



Perception Toolkits



Point Tracking
CoTracker3



Segmentation
SAM2



Geometric Fitting



ROI Photometry



Temporal Analysis

2 Generation Protocol

First-frame T2I

Example: Electromagnetic induction

Image prompt

A **bar magnet** on the axis of a **fixed copper coil**;
an **unlit indicator lamp** wired to the coil.

Front view, plain background, no battery.



GPT Image
T2I

First frame



Video prompt

Static camera, plain background.
Continue from the first frame.
.....(scene description).....

A hand takes hold of the end of the **bar magnet**, and pushes it steadily to the right **along the coil axis**.

Prompt Refinement

Generated video



T2V
VDM

3 Measurement Pipeline

Input video

Example: Pendulum oscillations



Temporal consistency **C** VLM check

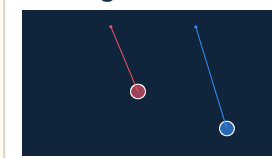
C ≥ 80

Programmatic physical measurement

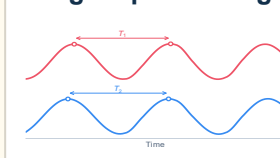
Tracking



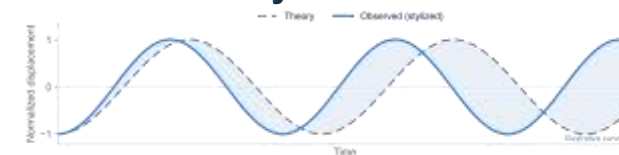
Segmentation



Signal processing



Observation vs. theory



M1 T-L relation

$$e = \left| \frac{(T_1/T_2)^2}{L_1/L_2} - 1 \right|$$

M2 Period stability

$$CV(T_j) = \frac{\sigma(T_j)}{\mu(T_j)}$$

Scale-invariant
Measurements

P = mean defined metric scores

$$\mathbf{S} = 0.15\mathbf{C} + 0.85\mathbf{P}$$



4 Overall Performance

Seedance
2.5

MiniMax
H3

Cosmos3
Super

VBVR
Wan2.2

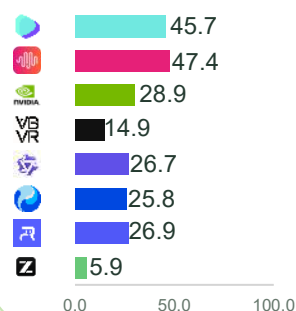
Wan2.2
I2V-A14B

Hunyuan Video
1.5

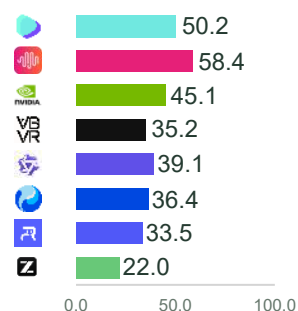
LingBot
30B-A3B

CogVideoX
1.5-5B

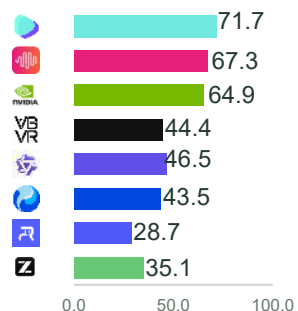
Translation & Collisions



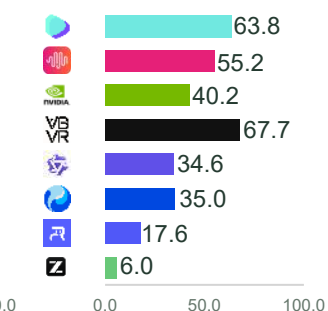
Rolling, Friction & Rigid Statics



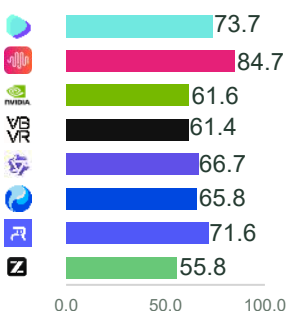
Pendulums & Oscillations



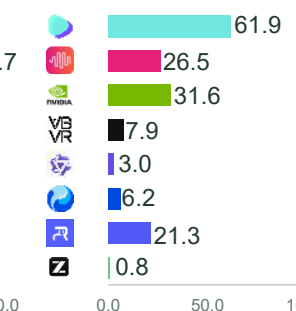
Optics & Projective Geometry



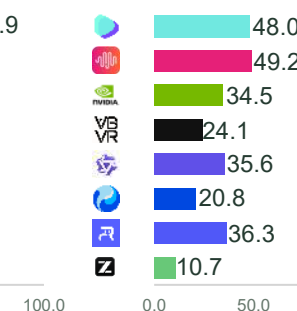
Hydrostatics & Buoyancy



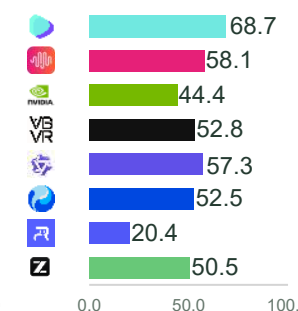
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